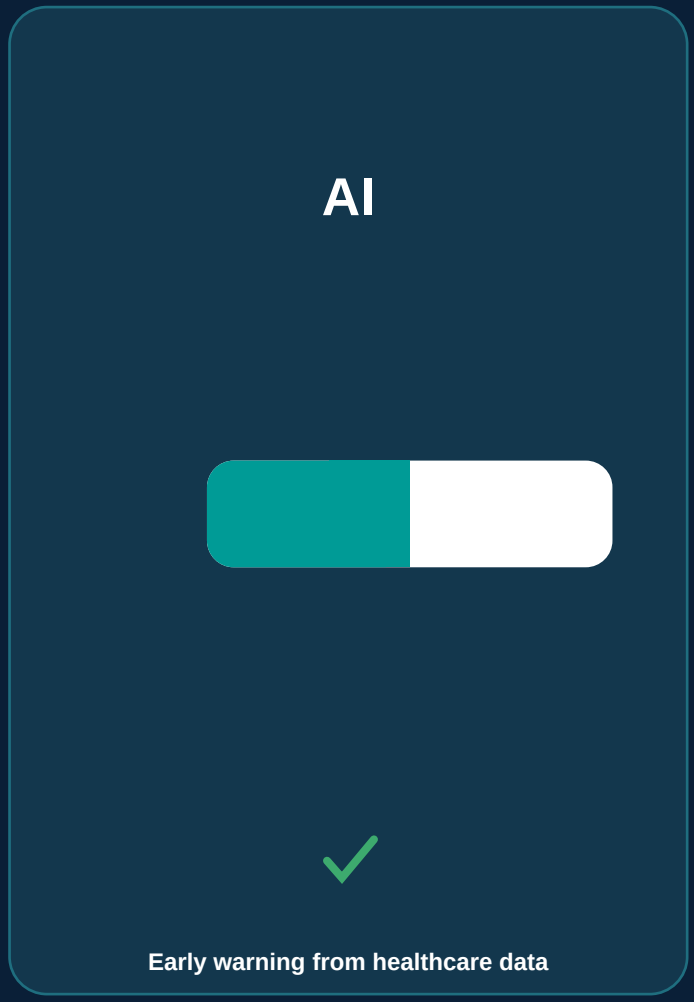
DRUG ADVERSE REACTION DETECTION USING HEALTHCARE DATA

Healthcare Informatics • Pharmacovigilance • Machine Learning



BRANCH

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01 • PROBLEM STATEMENT & OBJECTIVES

Why adverse drug reactions need intelligent detection, and what we aim to do

PATIENT → DRUG → REACTION

The challenge

- ADRs can be difficult to identify early.
- Healthcare records contain large volumes of clinical information.
- Manual pharmacovigilance is time-consuming and may miss hidden patterns.
- Patients can respond differently to the same medication.
- Machine Learning can prioritize patients and drug-event combinations.

Our objectives

01 — Collect & prepare
Turn healthcare data into a clean ML-ready dataset.

02 — Detect patterns
Find links between drugs, patient factors and adverse events.

03 — Predict risk
Use supervised ML to classify likely ADR outcomes.

04 — Support decisions
Show understandable risk signals to healthcare / PV teams.

Goal: turn healthcare data into an early-warning signal.

Key idea: predictive analytics should complement, not replace, clinical judgment.

THEORY 1 • ADVERSE DRUG REACTIONS & PHARMACOVIGILANCE

Basic ideas in simple words

What is an ADR?

A harmful, unintended response to a medicine taken at a normal dose for prevention, diagnosis or treatment (WHO definition, simplified).

Type A: Predictable — Augmented effect	
Cause	Extension of the drug's normal action
Dose	Usually dose-related
Frequency	Common, rarely fatal
Example	Bleeding with blood thinners

Type B: Unpredictable — Bizarre effect	
Cause	Not linked to normal action (e.g. allergy)
Dose	Usually not dose-related
Frequency	Uncommon, can be serious
Example	Severe penicillin allergy

What is pharmacovigilance?

The science and activity of detecting, assessing, understanding and preventing adverse effects or any other drug-related problem.

1. **Detect** — Spot a possible drug-reaction link
2. **Assess** — Check how likely and how serious
3. **Understand** — Find the cause and who is at risk
4. **Prevent** — Warn, relabel, guide safe use

Data: spontaneous reports

Doctors, pharmacists and patients report suspected reactions.

Data: health records

Diagnoses, lab results and prescriptions kept by hospitals.

Data: safety databases

National and global collections, e.g. WHO VigiBase, FDA FAERS, PvPI (India).

Why detect early? Some ADRs are mild (rash) but others are serious, so early detection lets doctors adjust the dose, switch drugs or monitor closely.

THEORY 2 • SIGNAL DETECTION & MACHINE LEARNING

Two ways to find possible drug-safety signals

Traditional: disproportionality

Looks at a database of reports and asks: is this reaction reported with this drug more often than with other drugs?

	Reaction Y	Other reactions
Drug X	**20**	180
All other drugs	100	9,700

Drug X: 20 of 200 reports (10%) mention reaction Y. Other drugs: 100 of 9,800 reports (about 1%).

10% versus 1% is about 10 times higher, so Y is a possible signal for human review.

Machine learning (supervised)

1. **Learn**
   Past records with known outcomes (ADR / no ADR)
2. **Find patterns**
   Links age, drug, dose and labs to outcomes
3. **Predict**
   A new patient gets an ADR risk score

Classification

Choosing a category: ADR or no ADR. Can also output a probability.

Features

The input clues: age, drug, dose, lab values, diagnosis.

Label

The correct answer in past data: did an ADR occur?

Train / test split

Learn on one part of the data, then test on cases never seen.

Together: statistics scan report databases for signals, while ML predicts risk from patient-level records. The model flags; a human decides.

Numbers in the example are illustrative.

LITERATURE REVIEW • PAPERS 1 TO 3

Easy-to-explain research papers related to our project

1 Predictive modeling of structured electronic health records for adverse drug event detection

Jing Zhao, Aron Henriksson, Lars Asker, Henrik Boström | BMC Medical Informatics and Decision Making, vol. 15 (Suppl. 4), S1 | BioMed Central (Springer Nature) | 25 Nov 2015

KEY FINDINGS
Trained random forests on hospital lab results and clinical codes. For some ADEs, using both beat codes alone.

EXPLAIN IT SIMPLY
Lab results plus diagnosis codes help a computer spot likely ADEs in patient records.

LINK TO OUR PROJECT
Same data type (structured EHR) and same kind of model (random forest) as our system.

2 Using Machine Learning for Pharmacovigilance: A Systematic Review

Patrick Pilipiec, Marcus Liwicki, András Bota | Pharmaceutics, vol. 14, issue 2, article 266 | MDPI | 2022 | DOI 10.3390/pharmaceutics14020266

KEY FINDINGS
Reviewed 16 studies using NLP on online user text. 14 of them reported positive results in finding ADRs.

EXPLAIN IT SIMPLY
People write about side effects online, and a computer can read it and pick out possible ADRs.

LINK TO OUR PROJECT
Our future scope: adding clinical text (NLP) to structured data.

3 Analyzing adverse drug reaction using statistical and machine learning methods: A systematic review

Yonsei University author team, South Korea | Medicine, vol. 101, no. 25, e29387 | Wolters Kluwer | June 2022

KEY FINDINGS
Reviewed 72 articles (2015 to 2020): 51 used statistics and 21 used ML. DrugBank was the most-used database for ML.

EXPLAIN IT SIMPLY
Older work counts reports with statistics; newer work uses ML on many data sources.

LINK TO OUR PROJECT
Explains why we use ML, and where our EMR and report data fit in.

LITERATURE REVIEW • PAPERS 4 & 5, AND REFERENCES

Easy-to-explain research papers related to our project

4 Predicting adverse drug event using machine learning based on electronic health records: a systematic review and meta-analysis

Qiaozhi Hu, Yuxian Chen, Dan Zou, Zhiyao He, Ting Xu (West China Hospital, Sichuan University) | Frontiers in Pharmacology, vol. 15, 1497397 | Frontiers Media | 13 Nov 2024

KEY FINDINGS

Reviewed 59 studies on ML for EHR-based ADE prediction. Random forest was used most, then SVM and XGBoost; average AUC about 77%.

EXPLAIN IT SIMPLY

Hospital records can predict who may get a side effect, and tree-based models lead.

LINK TO OUR PROJECT

Our features and models (random forest, XGBoost) match what this review found most useful.

5 Machine learning methods for predicting adverse drug events: A systematic review

Niaz Chalabianloo, Fatemeh Ahmadi, Mohammad Ali Omrani et al. | British Journal of Clinical Pharmacology | Wiley (British Pharmacological Society) | 2026 | DOI 10.1002/bcp.70377

KEY FINDINGS

59 studies, 191 ML runs. Logistic regression, random forest and XGBoost were most used. Only 20 of 59 handled class imbalance.

EXPLAIN IT SIMPLY

The popular models work well, but rare ADEs and testing on new hospitals need care.

LINK TO OUR PROJECT

Supports our model choices and our focus on recall, precision and F1.

References

[1] J. Zhao, A. Henriksson, L. Asker and H. Boström, “Predictive modeling of structured electronic health records for adverse drug event detection,” BMC Med. Inform. Decis. Mak., vol. 15, Suppl. 4, S1, 2015. doi:10.1186/1472-6947-15-S4-S1

[2] P. Pilipiec, M. Liwicki and A. Bota, “Using machine learning for pharmacovigilance: A systematic review,” Pharmaceutics, vol. 14, no. 2, p. 266, 2022. doi:10.3390/pharmaceutics14020266

[3] “Analyzing adverse drug reaction using statistical and machine learning methods: A systematic review,” Medicine (Baltimore), vol. 101, no. 25, e29387, Jun. 2022.

[4] Q. Hu, Y. Chen, D. Zou, Z. He and T. Xu, “Predicting adverse drug event using machine learning based on electronic health records: a systematic review and meta-analysis,” Front. Pharmacol., vol. 15, 1497397, 2024. doi:10.3389/fphar.2024.1497397

[5] N. Chalabianloo et al., “Machine learning methods for predicting adverse drug events: A systematic review,” Br. J. Clin. Pharmacol., 2026. doi:10.1002/bcp.70377

02 • DATA, FEATURES & PREPARATION

Turning healthcare records into clean, useful signals

What data we use

Patient
Age, sex, history, conditions

Medication
Drug, dose, frequency

Reaction
Symptoms, severity, event label

Context
Labs, diagnosis, encounter details

Illustrative record: 67-year-old male • Warfarin 5 mg daily • Reaction: bleeding (severe) • Lab: high INR

Preparing the data

1 Missing values
Fill blanks with a typical value or mark them as missing.

e.g. a lab test never done

2 Encoding
Convert categories such as sex or drug name into numbers.

e.g. Male / Female = 0 / 1

3 Class imbalance
ADRs are rare. Oversampling (SMOTE), undersampling or class weights help.

e.g. 1 ADR in 100 patients

4 Feature selection
Keep the most useful predictors and drop the noise.

e.g. keep age, dose, kidney labs

Avoid data leakage
Split into training and test sets first, and never let test cases influence how the model is trained.

Why it matters
A 2026 review found only 20 of 59 ADE prediction studies (33.9%) used SMOTE, cost-sensitive learning or stratified sampling.

03 • SYSTEM WORKFLOW & ARCHITECTURE

From raw healthcare data to an explainable ADR risk signal

1 2 3 4 5

Data Collection

Healthcare / pharmacovigilance records

Preprocessing

Clean, encode, handle missing values

Feature Engineering

Create meaningful clinical predictors

ML Model

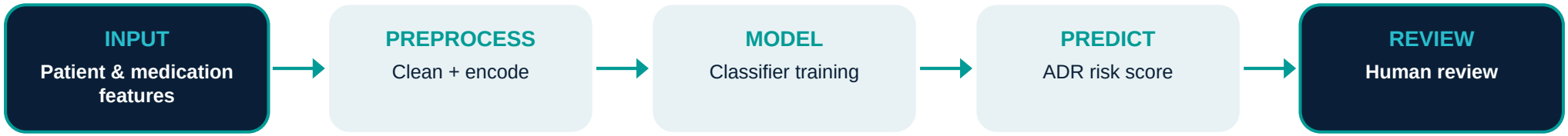
Train and validate classification model

Risk Output

ADR probability / predicted class

OUTPUT: “Higher ADR risk: flag for human review”

Model architecture



Explainability matters

Feature importance and model explanations help reviewers understand why a case was flagged.

Example: why a case is flagged

Age above 65 • high dose • abnormal lab value • takes several other drugs.

04 • MACHINE LEARNING APPROACH & EVALUATION

A practical classification pipeline and how we measure it

Logistic Regression

Strong baseline • interpretable

In simple words

Adds up weighted clues (age, dose, labs) into a probability from 0 to 1.

Decision Tree

Readable rules • feature splits

In simple words

A flowchart of yes/no questions, e.g. “Age above 65?” then “Dose high?”.

Random Forest

Robust ensemble • nonlinear patterns

In simple words

Many decision trees each give an answer, and the majority vote wins.

XGBoost / Boosting

High-performance candidate

In simple words

Trees are built one after another, each fixing the mistakes of the last.

How we measure the model

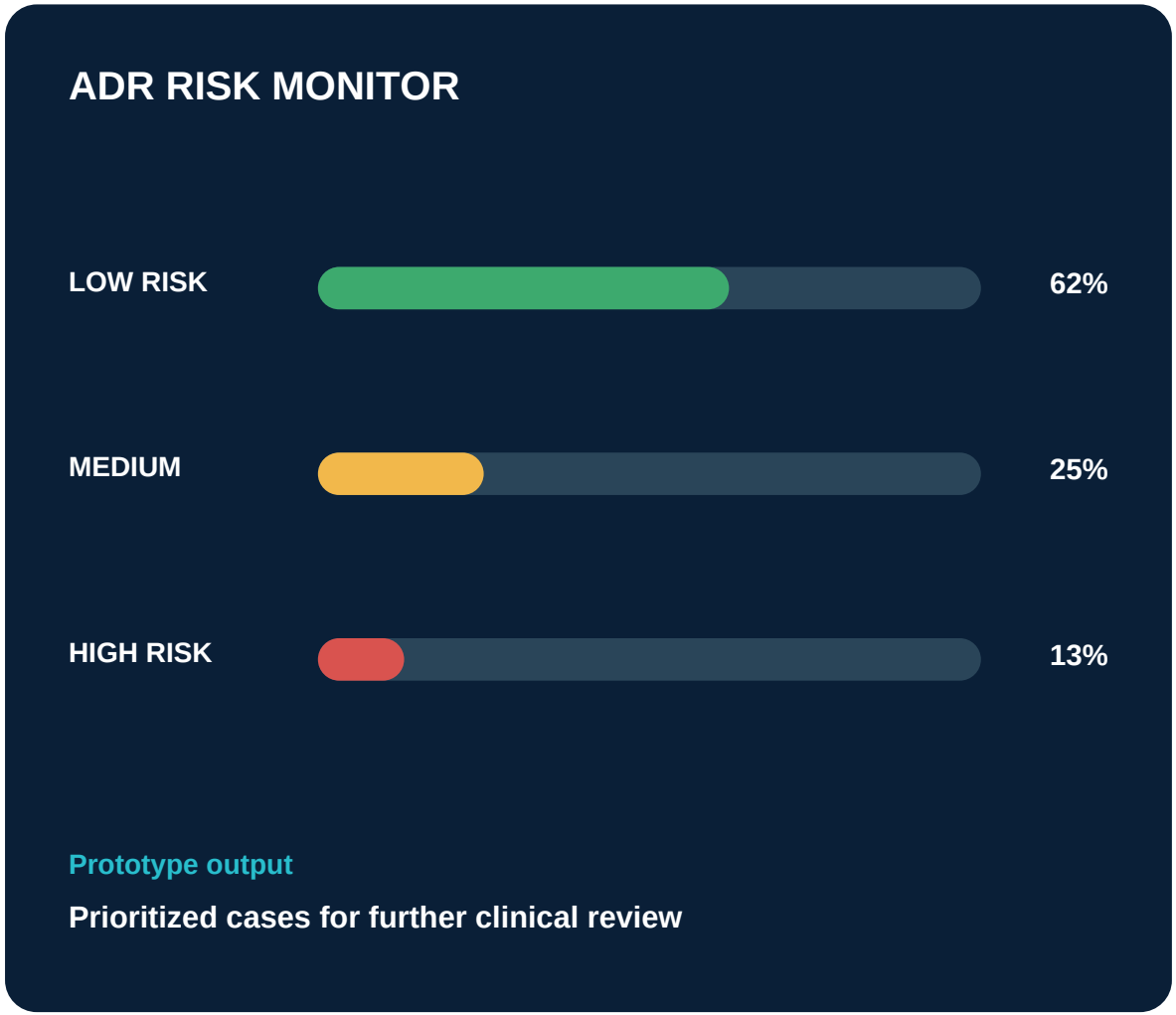
Metric	What it means	In ADR terms
✓ **Accuracy**	Share of all predictions that are correct.	Can look high even if rare ADRs are missed.
✓ **Precision**	Of the cases flagged as ADR, how many really are.	High precision means fewer false alarms for reviewers.
✓ **Recall**	Of all real ADR cases, how many the model found.	High recall means fewer missed reactions. Key for safety.
✓ **F1-score**	A balance of precision and recall in one number.	Useful when both false alarms and misses matter.
✓ **ROC-AUC**	How well the model ranks risky cases above safe ones (1.0 perfect, 0.5 guess).	Helps build a ranked review list.

Watch out: ADRs are rare

If only 1 in 100 patients has an ADR, a model that always says “no ADR” is 99% accurate but useless. Recall, precision and F1 matter more than accuracy alone.

05 • EXPECTED RESULTS

What a successful prototype should demonstrate



EARLIER

Surface signals sooner

SCALABLE

Process many records

CONSISTENT

Apply learned criteria

ACTIONABLE

Rank cases for review

High-risk cases rise to the top of the review list for the pharmacovigilance team.

Illustrative prototype output, not real patient data.

06 • APPLICATIONS, CHALLENGES & FUTURE SCOPE

Where the project can go, and what to watch out for

Applications

1

Hospital Safety

Monitor medication-related events across departments.

2

Pharmacovigilance

Prioritize suspected ADR reports for investigation.

3

Clinical Decision Support

Provide risk signals alongside patient context.

4

Population Analytics

Study patterns across drugs and patient groups.

5

Future AI

NLP on clinical notes, explainable AI, continuous learning.

Challenges, ethics & limitations

Data quality

Records can be missing or inconsistent. Poor data gives poor predictions.

Rare events

ADRs are uncommon, so models can miss them unless class imbalance is handled.

Generalisation

A model from one hospital may not work in another. Reviews note limited external validation.

Privacy

De-identify patient data and follow laws such as HIPAA (US) and DPDP Act 2023 (India).

Explainability & bias

Reviewers need reasons for each flag, and training data should represent all patient groups.

Human in the loop

The model only flags cases. Doctors and PV experts make the final decision.

Future direction: structured data + clinical text + explainable AI.

Responsible AI: protect patients, test on new data, and keep humans in charge.

07 • CONCLUSION

Safer medication use through data-driven detection.

DETECT

Find hidden patterns

PREDICT

Estimate ADR risk

SUPPORT

Help reviewers prioritize

THANK YOU

Questions & Discussion

MD Nayeem Uddin • Mohammed Junaid • Gulam Ishaq Siddiqui • MD Fahad Uddin

Guide: Ms. Ataliya Aarfeen | CSE 4th Yr / B2